

Walmart Retail Sales Dashboard

A Power BI Case Study

The Problem

- Walmart's weekly sales data spans 45 stores and many departments, but raw numbers alone do not show what needs attention
- Retailers need to know which stores are underperforming, how much holiday season really matters, and whether outside factors like fuel price or unemployment affect sales
- This project uses Walmart's public store sales dataset, covering three years of weekly sales, to answer these questions

The Approach

- Combined three data sources: weekly sales, store details, and external features like fuel price and unemployment
- Cleaned the data in Power Query, replacing missing values and standardizing store and date fields
- Built a proper data model with a date table to support time based calculations like year over year growth
- Designed four dashboard pages, moving from a high level summary down to store and department detail

The Solution

A four page interactive Power BI dashboard:

- **Executive Overview:** total sales, growth versus last year, holiday sales, number of stores, plus a weekly trend and store type breakdown
- **Store Performance:** every store ranked by total sales, with size, type, and average weekly sales
- **Seasonality and Holiday Impact:** holiday weeks highlighted on the trend line, holiday versus non holiday comparison, and sales against fuel price and unemployment
- **Department Breakdown:** top departments by sales, plus a heat map of department performance across store types

Key Insights

- Total sales across all 45 stores reached 6.74 billion dollars over three years
- Holiday weeks alone contributed 505.3 million dollars of that total
- Store Type A locations drove 4.3 billion dollars, more than double Store Type B and nearly ten times Store Type C
- Store 20 was the top performer at roughly 30.1 crore, closely followed by Store 4 and Store 14
- Holiday weeks showed a clear, repeating spike each year around Thanksgiving and Christmas
- Department 92 led all departments at 0.48 billion, followed by Department 95 at 0.45 billion, with the top two departments accounting for a large share of total department sales

Why This Matters

- Shows where attention should go next, not just what happened
- Flags which stores need investigation and which departments deserve more focus
- Gives a real sense of how much holiday planning is worth budgeting for
- Turns raw weekly transaction data into something a store manager or regional lead could actually use

Tools Used

Power BI, Power Query, DAX, Data Modeling



Executive
Overview



Store
Performance



Seasonality &
Holiday Impact



Department
Breakdown

6.74bn

Total Sales

42.22%

YoY Sales Growth

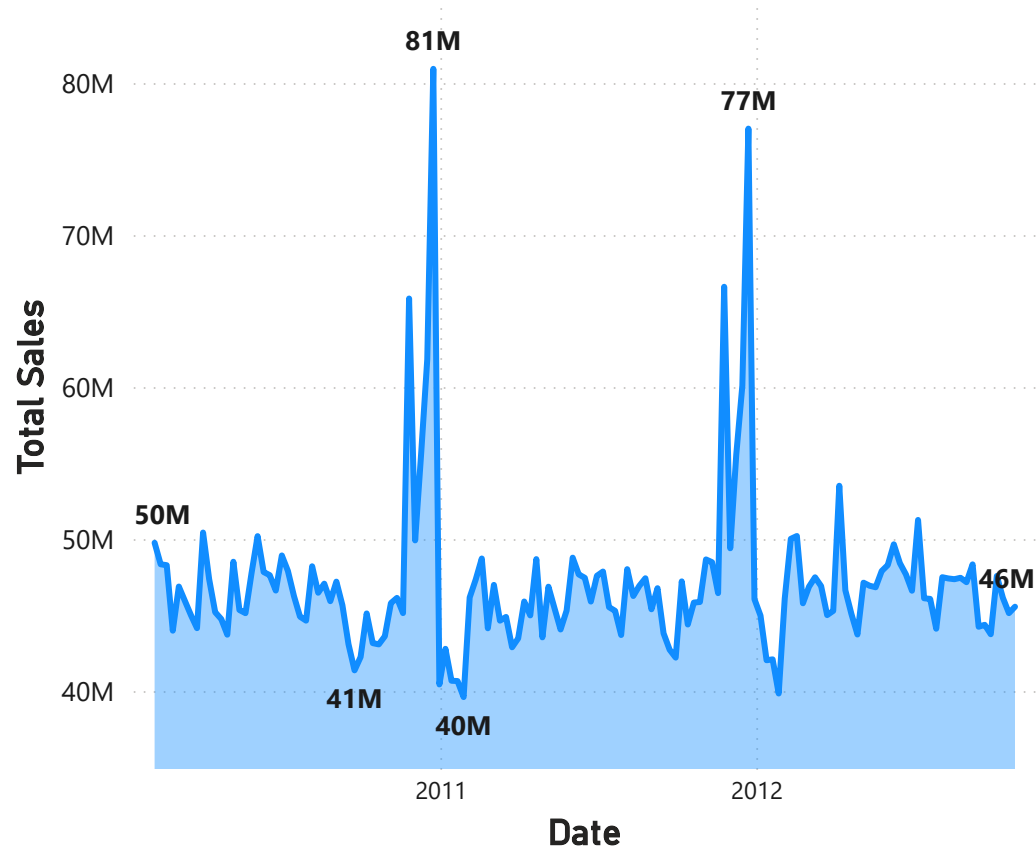
45

Number of Stores

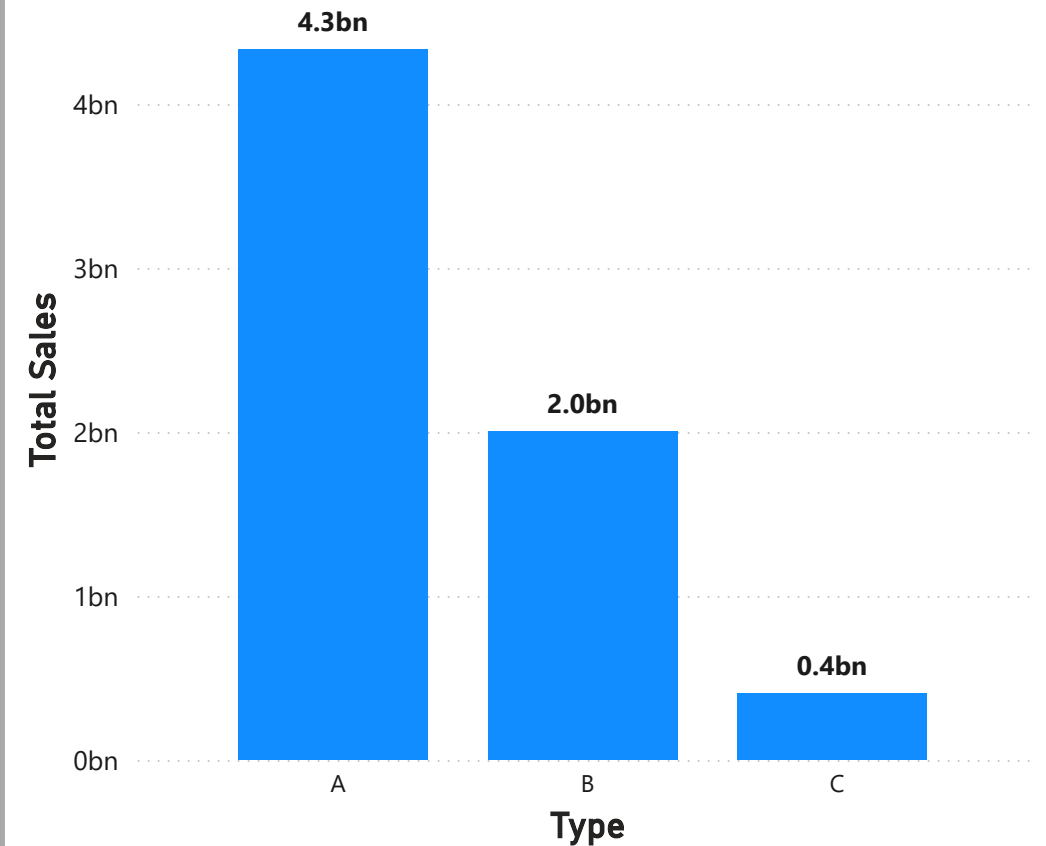
505.30M

Holiday Sales

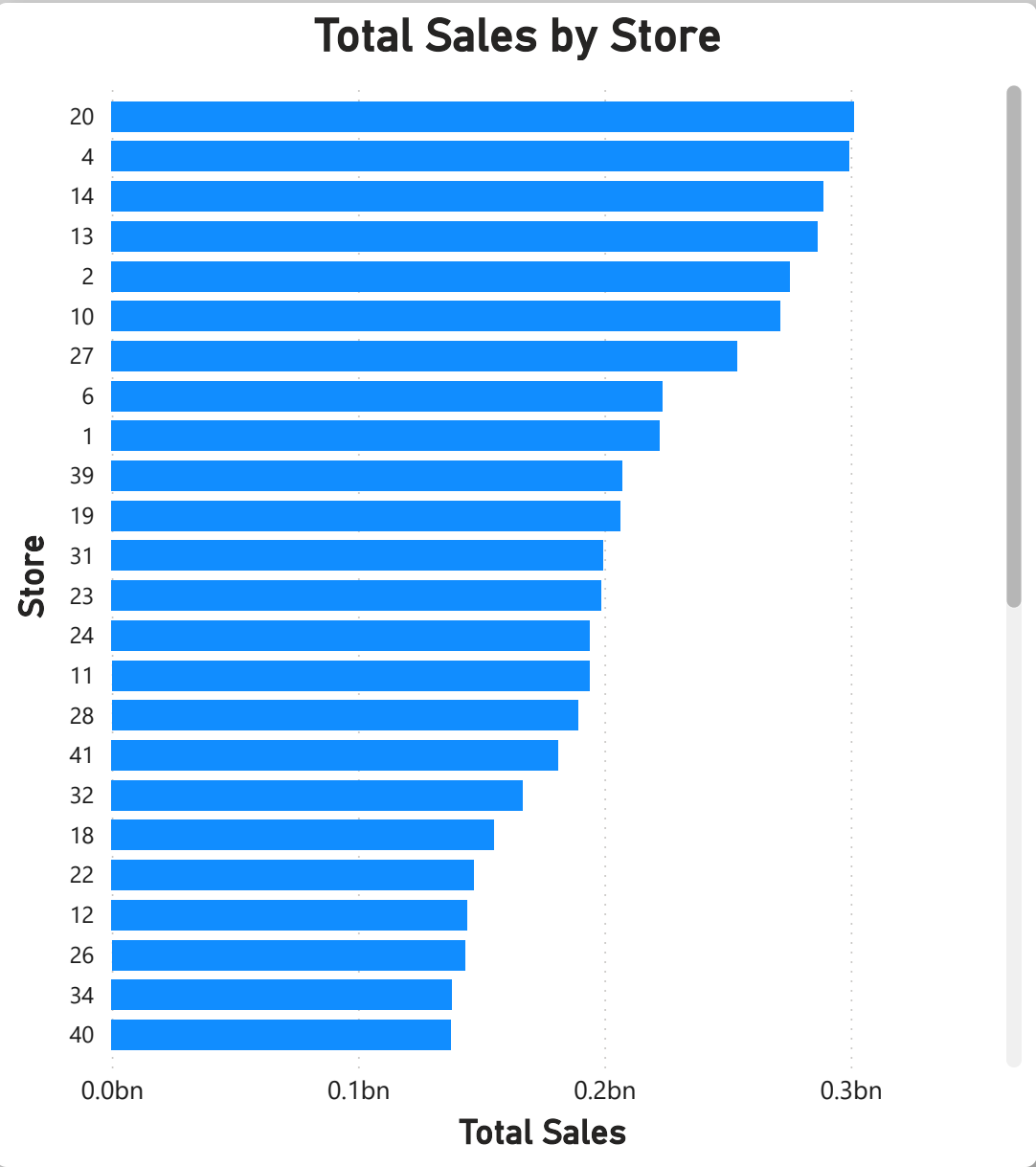
Weekly sales Trend



Sales by Store type



- Executive Overview
- Store Performance
- Seasonality & Holiday Impact
- Department Breakdown



Store Performance Details

Store	Type	Size	Total Sales	Avg Weekly Sales
20	A	203742	30,13,97,792.46	21,07,676.87
4	A	205863	29,95,43,953.38	20,94,712.96
14	A	200898	28,89,99,911.34	20,20,978.40
13	A	219622	28,65,17,703.80	20,03,620.31
2	A	202307	27,53,82,440.98	19,25,751.34
10	B	126512	27,16,17,713.89	18,99,424.57
27	A	204184	25,38,55,916.88	17,75,216.20
6	A	202505	22,37,56,130.64	15,64,728.19
1	A	151315	22,24,02,808.85	15,55,264.40
39	A	184109	20,74,45,542.47	14,50,668.13
19	A	203819	20,66,34,862.10	14,44,999.04
31	A	203750	19,96,13,905.50	13,95,901.44
23	B	114533	19,87,50,617.85	13,89,864.46
24	A	203819	19,40,16,021.28	13,56,755.39
11	A	207499	19,39,62,786.80	13,56,383.12
28	A	206302	18,92,63,680.58	13,23,522.24
41	A	196321	18,13,41,934.89	12,68,125.42
32	A	203007	16,68,19,246.16	11,66,568.15
Total			6,73,72,18,987.11	4,71,13,419.49



Executive
Overview



Store
Performance

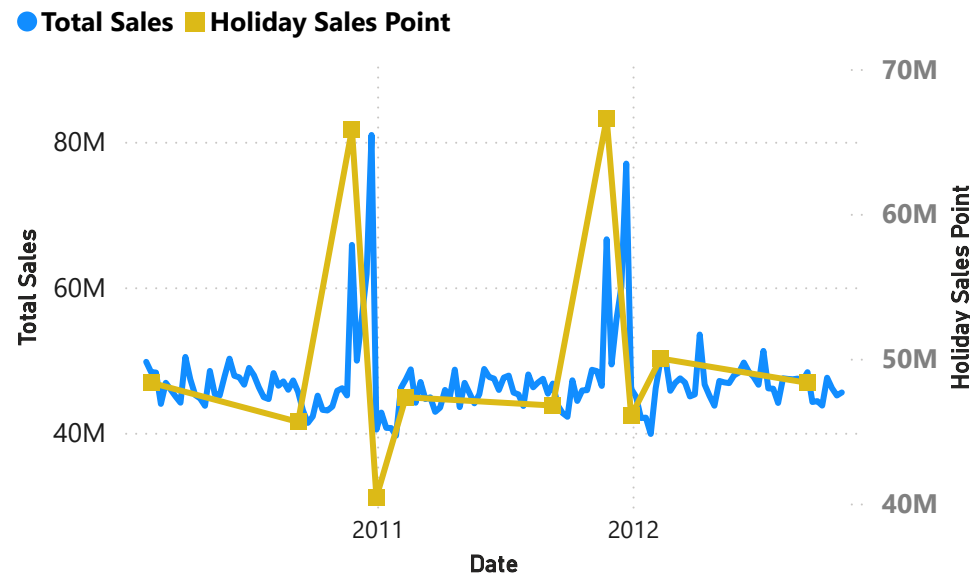


Seasonality &
Holiday Impact

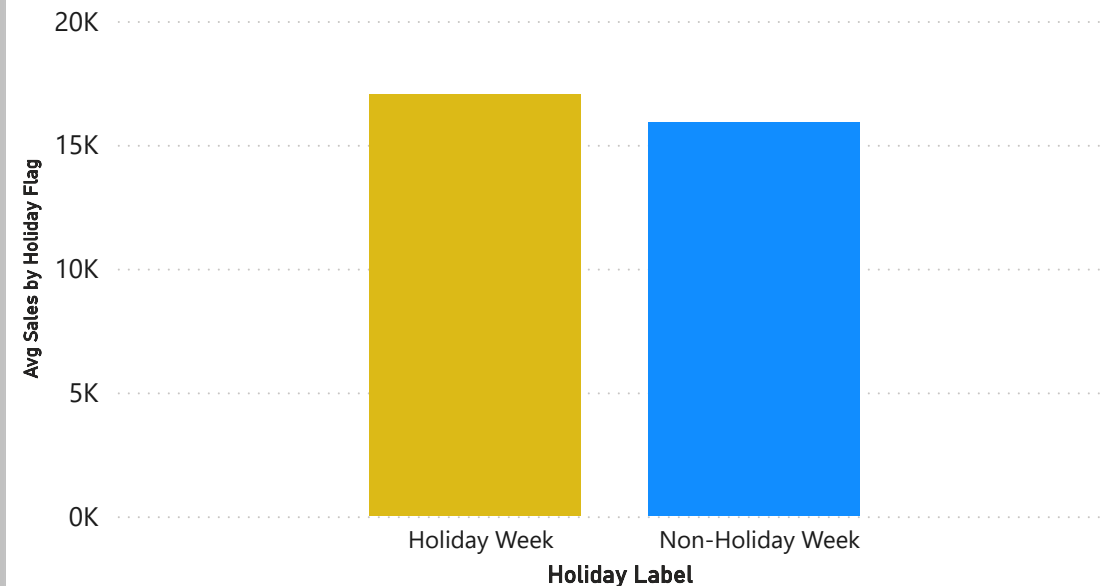


Department
Breakdown

Weekly Sales with Holiday Weeks Highlighted



Average Sales: Holiday vs Non-Holiday Weeks



Sales Trend vs Fuel Price and Unemployment Rate



Date 

01-01-2010 

31-12-2012 





Executive
Overview



Store
Performance



Seasonality &
Holiday Impact

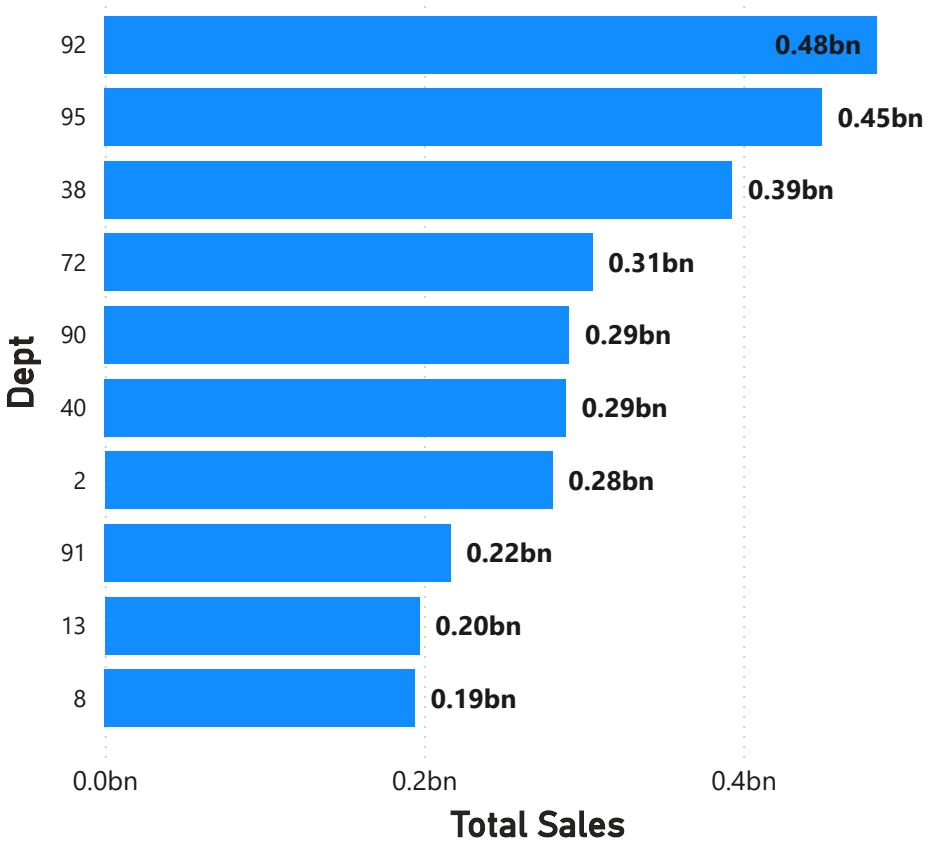


Department
Breakdown

Department Sales by Store Type

Dept	A	B	C	Total
92	35,28,45,549.71	7,89,35,030.62	5,21,62,761.54	48,39,43,341.87
95	30,54,57,805.93	10,04,11,893.93	4,34,50,462.66	44,93,20,162.52
38	20,45,05,786.91	14,17,30,138.72	4,68,82,211.29	39,31,18,136.92
72	17,85,08,245.74	12,71,27,225.80	89,680.67	30,57,25,152.21
90	22,19,51,879.82	3,16,83,561.60	3,74,33,022.26	29,10,68,463.68
40	16,87,67,879.76	10,29,90,667.89	1,71,77,474.40	28,89,36,022.05
2	16,35,75,247.15	10,46,59,404.51	1,23,76,522.77	28,06,11,174.43
91	16,90,47,601.60	2,14,58,812.95	2,62,75,291.18	21,67,81,705.73
13	11,77,26,331.69	6,83,87,387.53	1,12,07,850.73	19,73,21,569.95
8	11,66,88,981.25	6,70,44,326.37	1,05,47,473.11	19,42,80,780.73
94	16,06,56,930.21	21,11,682.06	2,71,43,838.06	18,99,12,450.33
4	10,37,35,619.08	5,16,82,806.70	1,17,28,319.80	16,71,46,745.58
93	13,62,10,993.40	30,44,717.29	2,04,42,952.50	15,96,98,663.19
7	9,68,53,926.40	5,81,39,150.78	4,84,485.57	15,54,77,562.75
79	9,03,83,435.55	4,33,14,113.30	76,98,995.29	14,13,96,544.14
23	7,49,15,762.05	6,32,49,773.27	14,795.46	13,81,80,330.78
5	8,34,92,740.66	5,14,99,769.69	6,14,848.22	13,56,07,358.57
9	7,36,85,010.90	5,46,09,999.02	98,246.73	12,83,93,256.65
46	7,42,85,803.26	4,84,46,291.26	56,12,316.92	12,83,44,411.44
1	7,22,22,369.29	4,37,35,819.94	76,80,587.31	12,36,38,776.54
10	7,47,42,454.56	4,28,66,796.33	2,88,136.69	11,78,97,387.58
81	7,08,10,850.34	1,79,90,264.28	1,07,50,914.69	9,95,52,029.31
82	6,27,29,856.41	3,17,77,650.97	50,39,572.95	9,95,47,080.33
14	5,58,06,268.62	3,88,46,827.73	10,41,570.10	9,56,94,666.45
11	5,39,40,995.04	3,86,06,281.28	7,82,000.06	9,33,29,276.38
16	5,77,94,471.48	3,25,39,005.07	13,37,205.72	9,16,70,682.27
97	6,95,07,117.14	84,75,427.03	1,15,13,967.73	8,94,96,511.90
Total	4,33,10,14,722.75	2,00,07,00,736.82	40,55,03,527.54	6,73,72,18,987.11

Top 10 Departments by Sales



Store

Dept